Lines and Planes in Space

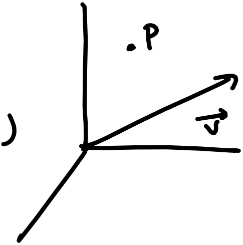
Section 11.5

8-26-26

A) Equation of a line in space

$\parallel$ to $\vec{v} = \langle a, b, c \rangle$

through $P(x, y, z)$



Parametric Form

$x = x_1 + at \qquad t = \dfrac{x - x_1}{a}$

$y = y_1 + bt \qquad t = \dfrac{y - y_1}{b}$

$z = z_1 + ct \qquad t = \dfrac{z - z_1}{c}$

Symmetric Form

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$$

IMPORTANT: T values don't have to be the same

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ex) through $(5,1,3)$ // $\langle 1,2,-1\rangle$

Parametric:

$x = 5 + t$

$y = 1 + 2t$

$z = 3 - t$

2 more points

$(6,3,2)$

$(4,-1,4)$

Symmetric:

$$\frac{x-5}{1}=\frac{y-1}{2}=\frac{z-3}{-1}$$

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ex) Through $A(2,4,-3)$ and $B(3,-1,1)$

$$\overrightarrow{AB}=\langle 1,-5,4\rangle$$

$x = 2 + t$

$y = 4 - 5t$

or

$x = 3 + t$

$y = -1 - 5t$

$z = -3 + 4t$ $\qquad$ $z = 1 + 4t$

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ex) Show if the lines are skew

$x = 1 + t$ $\qquad$ $x = 2s$

$y = -2 + 3t$ $\qquad$ $y = 3 + s$

$z = 4 - t$ $\qquad$ $z = -3 + 4s$

$\langle 1, 3, -1 \rangle$ $\qquad$ $\langle 2, 1, 4 \rangle$

not $//$

$2s = 1 + t$

$3 + s = -2 + 3t$

not $\perp$

$t = \frac{11}{5}$

$s = \frac{8}{5}$

If they intersect, Z should be the same

$z = 4 - \frac{11}{5}$ $\qquad$ $z = -3 + \frac{32}{5}$

not intersecting, they are skew

B) Planes

contains point $P(x, y, z)$

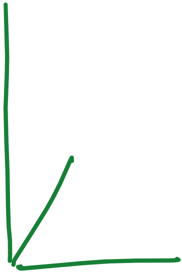
and $\vec{N} = \langle a, b, c \rangle$

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0$$

$$ax+by+cz$$

ex) Find the Plane containing

$P(1,0,2)$ $Q(3,-1,6)$ $R(5,2,4)$



$\overrightarrow{PQ} = \langle 2, -1, 4\rangle$

$\overrightarrow{PR} = \langle 4, 2, 2\rangle$

$\overrightarrow{PQ} \times \overrightarrow{PR} = \vec{N} = \langle -10, 12, 8\rangle$

$\langle -5, 6, 4\rangle$

$-5x + 6y + 4z$

$= 3$

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3) a) Parametric

$x = 3t \quad y = t \quad z = 5t$

b) Symmetric

$$\frac{x}{3} = \frac{y}{1} = \frac{z}{5}$$

21) Pt: $(3,-1,-2)$ $\vec{v} = \langle -1,2,0 \rangle$

25) $L_1: \langle -3,2,4 \rangle$ $L_1 = L_2$

$L_2: \langle 6,-4,-8 \rangle$ $L_1 = L_2 \parallel L_3$

$L_3: \langle -6,4,8 \rangle$

~~$L_4: \langle 6,4,-6 \rangle$~~ not identical

27) $L_1: \langle 4,-2,3 \rangle$ $L_1 = L_3$

~~$L_2: \langle 2,1,5 \rangle$~~

$L_3: \langle -8,4,-6 \rangle$

~~$L_4: \langle -2,1,1.5 \rangle$~~

29) x: $4t+2 = 2s+2$ $s=0$

y: $3 = 2s+3$ $t=0$

z: $-t+1 = s+1$

sub s, t into z equation, lines intersect

$P(2,3,1)$

$u = 4i - k$

$v = 2i + 2j + k$

$$\cos\theta = \frac{|u \cdot v|}{\|u\|\|v\|} = \frac{8-1}{\sqrt{17}\sqrt{9}} = \frac{7}{3\sqrt{17}} = \frac{7\sqrt{17}}{51}$$

35) $(1,3,-7)$ $n=j$ $\langle 0,1,0\rangle$

$$0(x-1)+1(y-3)+0(z-(-7)) = 0$$

$$y-3 = 0$$

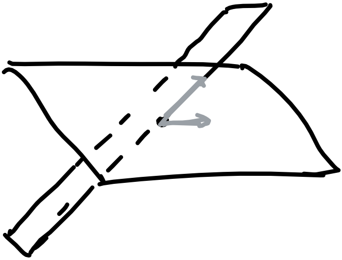
43) $u=2i-2k$ $v= -2i-4j-k$

$$\left(\frac{1}{2}u\right)\times(-v) = \begin{vmatrix} i & j & k \\ 1 & 0 & -1 \\ 2 & 4 & 1 \end{vmatrix} = 4i-3j+4k$$

$$4(x-1)-3(y-2)+4(z-3)=0$$

$$4x-3y+4z=10$$

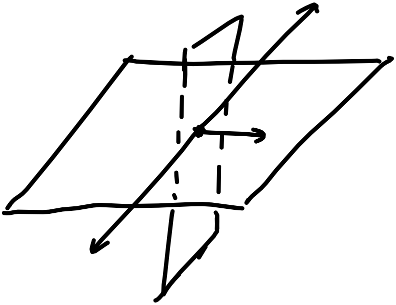
C) Angle Between 2 Planes



∡ between 2 planes is ∡ between normal vectors $\vec{n}_1$ and $\vec{n}_2$

$$\cos\theta = \frac{|n_1 \cdot n_2|}{\|n_1\|\|n_2\|}$$

D) Finding the Line of Two Intersecting Planes



we need a point on line direction vector of line.

$\vec{n}_1 \times \vec{n}_2$

ex) $x+y+z=1$

$x-2y+3z=1$

$n_1 = \langle 1, 1, 1 \rangle$

$n_2 = \langle 1, -2, 3 \rangle$

$\vec{n_1} \times \vec{n_2} = \langle 5, -2, -3 \rangle$

find a point that satisfies both equations

$(1, 0, 0)$

Parametric

$x = 1+5t$

$y = -2t$

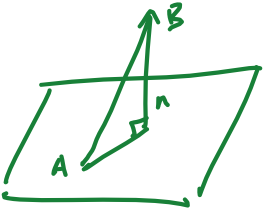
$z = -3t$

Symmetric

$$\frac{x-1}{5} = \frac{y}{-2} = \frac{z}{-3}$$

E) Distance from point to plane

Ex) Project $\vec{AB}$ onto $\vec{n}$



$A(0,1,0)$

$B(2,3,4)$

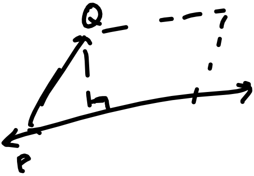
$\vec{AB} = \langle 2,2,4\rangle$

$\vec{n} = \langle 5,1,-1\rangle$

$\vec{AB}\cdot\vec{n} = 8$

$$\frac{\vec{AB}\cdot\vec{n}}{\|\vec{n}\|^2}\cdot\vec{n} = \frac{8}{\sqrt{27}}$$

F) Distance between point and line



find point on line ?

$\vec{PQ}$

$\|\vec{PQ}\times\vec{u}\|$

$$h = \frac{\|\vec{PQ} \times \vec{u}\|}{\|\vec{u}\|}$$

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14) $\langle 0, 1, 0 \rangle$

$x = 0, \ y = t, \ z = 0$

15) $(2, 3, 4)$

$v = 3i + 2j - k$

$x = 2 + 3t, \ y = 3 + 2t, \ z = 4 - t$

17) $(5, -3, -4)$

$v = \langle 2, -1, 3 \rangle$

$x = 5 + 2t, \ y = -3 - t, \ z = -4 + 3t$

19) $(2, 1, 2)$

$v = \langle -1, 1, 1 \rangle$

$x = 2 - t, \ y = 1 + t, \ z = 2 + t$

51)

$\vec{AB} = \langle 1, 7, 7 \rangle$

$\vec{u} = \langle 1, 0, 0 \rangle$

$\vec{n} = \vec{AB} \times u = \langle 0, 7, -7 \rangle$

$\vec{n} = \langle 0, 1, -1 \rangle$

71)

$$\left.\begin{aligned} P_1 &: -5x + 2y - 8z = 6 \\ P_2 &: 15x - 6y + 24z = 17 \end{aligned}\right\} \text{parallel}$$

$P_3: 6x - 4y + 4z = 9$

$P_4: 3x - 2y - 2z = 4$

$(0, -1, -1)$ on plane 1

$(0, -1, -1)$ not on plane 2

75) a) $n_1 = \langle 3, 2, -1 \rangle$

$n_2 = \langle 1, -4, 2 \rangle$

$$\cos\theta = \frac{7}{\sqrt{14} \cdot \sqrt{21}}$$

b) $n_1 \times n_2 = \langle 0, -7, -14 \rangle$

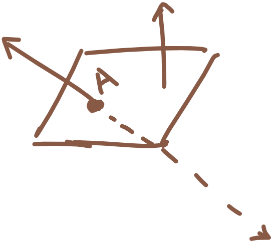
$\langle 0, 1, 2 \rangle$

77) $x = \frac{1}{2} + t$

$y = -3/2 - t$

$z = -1 + 2t$

$t = 3/2$



$(2, -3, 2)$

line is not in the plane

83) $Q(2, 8, 4)$

Plane: $2x + y + z = 5$

$n = \langle 2, 1, 1 \rangle$

$P(0, 0, 5)$

$$\overrightarrow{PQ} = \frac{11}{\sqrt{6}} = \frac{11\sqrt{6}}{6}$$

87) $n_1 = \langle -3, 6, 7\rangle$

$n_2 = \langle 6, -12, -14\rangle$

$n_2 = -2n_1$ planes are parallel

$P(0,-1,1)$ exists in $-3x+6y+7z=1$

$Q(\frac{25}{6}, 0, 0)$ exists in $6x-12y-14z=25$

$$\overrightarrow{PQ} = \left\langle \frac{25}{6}, 1, -1 \right\rangle$$

$$D = \frac{|\overrightarrow{PQ} \cdot n_1|}{\|n_1\|} = \frac{|-27/2|}{\sqrt{94}} = \frac{27}{2\sqrt{94}}$$

89) $v = \langle 4, 0, -1\rangle$

$Q(1,5,-2)$

$P(-2, 3, 1)$

$\overrightarrow{PQ} = \langle 3, 2, -3 \rangle$

$$\overrightarrow{PQ} \times v \begin{vmatrix} i & j & k \\ 3 & 2 & -3 \\ 4 & 0 & -1 \end{vmatrix} = \langle -2, -9, -8 \rangle$$

$$D = \frac{\|\overrightarrow{PQ} \times v\|}{\|v\|} = \frac{\sqrt{149}}{\sqrt{17}}$$

93) L_1 dir. vector $v_1 = \langle -1, 2, 1 \rangle$

L_2 dir. vector $v_2 = \langle 3, -6, -3 \rangle$

$v_2 = -3v$, lines are parallel

$Q(2, 3, 4)$ on L_1

$P(0, 1, 4)$ on L_2

$\overrightarrow{PQ} = \langle 2, 2, 0 \rangle$

$\sqrt{22/3}$

$v = v_2$

$$\vec{PQ} \times v_2 = \begin{vmatrix} i & j & K \\ 2 & 2 & 0 \\ 3 & -6 & -3 \end{vmatrix} = \langle -6, 6, -18 \rangle$$

$$D = \frac{\| \vec{PQ} \times v_2 \|}{\| v_2 \|} = \frac{\sqrt{36+36+324}}{\sqrt{9+36+9}}$$

$$= \frac{\sqrt{396}}{\sqrt{54}} = \sqrt{22/3}$$